ADITYA BAJAJ

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EDUCATION

University of Maryland, College Park *Expected Fall 2027*
B.S. Mechanical Engineering

- **Relevant Coursework**: Assistive Robotics, Mechatronics and Internet of Things, Electronics and Instrumentation II, Machine Design, Control Systems, Applied Machine Learning, Product Engineering and Manufacturing.

TECHNICAL EXPERIENCE

Vacuum System Redesign *August 2026 - Present*
Project Manager

- Leading a team of 6 engineering students to redesign and improve a commercially designed vacuum cleaner.
- Directing multi-system design efforts including suction performance, energy efficiency, thermal management, power consumption, and sustainability at minimum manufacturing cost.
- Coordinating team members, establishing project priorities, delegating responsibilities, tracking progress, and integrating individual design efforts into a final system deliverable.

6 DOF Robotic Arm *Sep 2025 - Dec 2025*
Private Project Endeavor

- Built and integrated a multi-degree-of-freedom robotic arm assembly with 43 3D-printed components and commercial hardware.
- Performed tolerancing on 3D-printed parts over 4 iterations to accommodate commercial hardware from various vendors and account for non-standard dimensions.
- Designed wiring channels within linkages to neatly organize and later identify wiring, which was soldered and crimped for desired dimensions, for servo components.
- Diagnosed and resolved potentiometer input discrepancies within 4 percent of error.

Material Identification Rover *Jan 2024 - May 2024*
Systems and Navigation Lead

- Programmed navigation in Arduino IDE which utilized force sensors to accurately identify the weight and material properties of cubic matter.
- Calculated and specced out motors for optimal traction, powder, and speed of an autonomous vehicle.
- Corrected motor and Arduino board interfacing to sync Aruco marker tracing and streamline navigation process to under 5 minutes.

Cube Satellite Team *Sep 2023 - Feb 2024*
CAD and Structures Development Member

- Helped develop a student CubeSat platform with a GPS tracking payload as a part of NASA mission.
- Engaged in FEA to estimate performance prior to fabrication and simulate duress under 10G load.
- Utilized Siemens NX to optimize structural supports and systems of Cube Satellite, optimizing the functionality of onboard electronics and reducing overall weight.

TECHNICAL SKILLS:

Programming: Python, MATLAB, C++, Arduino IDE, and Processing.
CAD and Simulation: SolidWorks, Siemens NX, ANSYS, Fusion360, QIDI Studio, Tinker CAD, and Multisim.
Manufacturing: 3D-printing, Wiring, Soldering, Crimping, Woodworking, Machine Maintenance, Computer Assembly, and Laser Cutting.